

BENDING OF BEAM

The apparatus consists essentially of a beam of the material under test about 1 metre long and of uniform rectangular cross section supported on two horizontal knife edges fixed parallel to each other in the same level on a rigid support. A hanger is attached to the middle point of the bar by means of a rectangular hook. Slotted weights can be placed on this hanger to load the beam in the middle. Just above the middle point of the beam is supported a spherometer in a frame fixed on a rigid support. When an electric circuit containing Galvanometer is properly arranged, the galvanometer shows a deflection as the screw of spherometer comes in contact with the beam refer to Fig. (1)

EXPERIMENT 1 :

To determine the value of Young's modulus for the material of the beam by the method of bending of beam.

APPARATUS : A uniform rectangular bar of iron about one meter long, 2.5cm wide, & 0.5cm thick, Two knife edge fixed on rigid support, A stirrup with a hook, hanger and Slotted Weights, Metre Rod, Vernier Calliper, Screw Gauge, and a Spherometer fitted on cast iron stand with electrical fitting.

FORMULA USED :

$$\text{Young Modulus } y = \frac{Mgl^3}{4\delta bd^3}$$

l = length of the beam between the two knife edges

b = breadth of the beam, d = thickness of the beam,

δ = depression at the middle point of the beam

PROCEDURE:

1. Place the experimental bar symmetrically on the two knife edges with equal length projecting beyond the knife edges.
2. Measure the distance between knife edges with a metre scale. It gives the length (l) of the beam.
3. Suspend the hanger from the stirrup resting on the bar midway between the two knife edges.
4. Arrange the spherometer.
5. Determine pitch and least count of the spherometer.



spherometer.

11. Find out the depression δ for a given difference in load.
12. Value of Young modules (y) can also be determined by the method of graph.

OBSERVATIONS:

[A] Measurement of dimensions of the beam

- (i) Length of the beam between the two knife edges $l = \dots$ cm
- (ii) Measurement of breadth b and thickness d of the beam.
 - (a) Vernier constant $= \dots$ cm
Zero error $= \dots$ cm
 - (b) Pitch of the screw $= \dots$ mm
No. of divisions on the circular scale $= \dots$ mm
 $\therefore$ Least count $= \dots$ mm
Zero error $= +$ or $- \dots$ mm

Sr.No	Breadth (cm.)	Mean breadth b (cm)	Thickness (mm.)	Thickness (mm)
1				
2				
3				

Corrected breadth (b) ..cms, corrected thickness (d) ...mm = cms

[B] Measurement of depression δ of the beam

Sr. No.	Load on the Hanger (gm)	Reading of the spherometer		Mean Reading (mm.)	Depression δ for 2000gm (mm.)
		Load	Load		
		Increasing (mm.)	Decreasing (mm.)		
1.	0				
2.	500				
3.	1000				
4.	1500				
5.	2000				
6.	2500				

Mean (δ) for 2 kg. weight

CALCULATIONS :

From table [B].

$\delta = \dots\dots\dots$ mm. = $\dots\dots\dots$ cm. for $M = \dots\dots\dots$ gm.

$b = \dots\dots\dots$ cm. $d = \dots\dots\dots$ cm.

$l = \dots\dots\dots$ cm. $g = \dots\dots\dots$ cm./sec²

$$\text{Hence } y = \frac{Mgl^3}{4bd^3\delta}$$

$$= \dots\dots\dots$$

$$= \dots\dots\dots \text{ dynes/cm}^2.$$

Alternatively, plot a graph between load (**M**) on **X-axis** and mean spherometer reading (δ) on **Y-axis** as shown in Fig 3 slope of the graph gives average value of

$$\delta = \tan \theta = \frac{p}{q}$$

$$\frac{M}{\text{and (Y) can be calculated.}}$$

PRECAUTIONS :

1. The beam must be placed symmetrically on the knife edges with equal length projecting beyond the two knife edges.
2. The stirrup must be placed with its knife edge exactly midway between the two knife edges. The surface of the beam where the spherometer screw touches, it must be clean.
3. The weights should be placed or removed from the hanger very gently.
4. Thickness of the beam should be measured very accurately using screw Gauge.
5. The maximum load applied should be within elastic limit.



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